STUDENT MANAGEMENT MODULE (CORE ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.3 · March 2026
Confidential · Internal Documentation

Audience: School Administrators · Teachers · IT Team · Parents

Related Modules:
Admissions · Academics · Attendance · Behavior · Health · Finance · Communication · Parent Portal

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1. MODULE OVERVIEW

1.1 Purpose

The **Student Management Module** serves as the **central system of record** for all student data, managing the full lifecycle:

```
Admission → Enrollment → Academic Progress → Behavior → Graduation → Alumni
```

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2. CORE OBJECTIVES

1. Maintain **complete student profiles**
2. Enable **real-time tracking (attendance, academics, behavior)**
3. Support **parent and teacher interaction**
4. Provide **actionable analytics and alerts**
5. Ensure **data security and role-based access**
6. Integrate seamlessly with all other modules

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3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

```
Applicant → Admitted → Active Student → Monitored → Promoted → Graduated →
Alumni
```

3.2 Daily Operational Flow

```
Student attends class
  ↓
Teacher marks attendance
  ↓
System updates attendance stats
  ↓
Grades entered after assessments
  ↓
Behavior incidents logged (if any)
  ↓
Parent notified (if triggered)
```

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4. MODULE ARCHITECTURE

```
Student Module
│
├── Student Profiles
├── Admissions Integration
├── Academic Records
├── Attendance Tracking
├── Behavior Management
├── Health Records
├── Document Management
├── Parent Linking
├── Communication Integration
├── Analytics & Reports
├── Alerts & Notifications
```

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5. STUDENT PROFILES (CORE DATA LAYER)

5.1 Features

Each student must have a **complete digital profile**:

- Personal details (name, DOB, gender)
- Admission number
- Class/grade
- Parent/guardian details
- Medical info
- Academic history
- Behavior records

5.2 Data Model

```
Student
- student_id
- admission_number
- full_name
- dob
- gender
- class_id
- parent_id
- status
- enrollment_date
```

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6. ADMISSIONS INTEGRATION

6.1 Automation

```
Student accepted in Admissions Module
  ↓
Student record auto-created
  ↓
Assigned admission number
  ↓
Linked to class and parent
```

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7. ACADEMIC RECORDS (CRITICAL GAP FIX)

7.1 Features

- Subject-level grades
- Exam results
- Report cards
- Teacher comments
- Performance trends

7.2 Real-World View

Instead of just “no data”, system shows:

```
Math: A
English: B+
Science: A-
Trend: Improving
```

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8. ATTENDANCE TRACKING (MAJOR UPGRADE)

8.1 Features

- Daily roll-call
- Late arrivals
- Absences with reasons
- Attendance trends

8.2 Automation Options

- Manual teacher entry
- QR code scanning
- Biometric systems
- Timetable-based auto prompts

8.3 Rule Engine

```
IF attendance < 85%
  → Trigger alert
  → Notify parent
```

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9. BEHAVIOR MANAGEMENT

9.1 Features

- Incident logging
- Categorization:
  - Minor
  - Moderate
  - Severe
- Resolution tracking
- Disciplinary actions

9.2 Data Model

```
Behavior
- incident_id
- student_id
- type
- severity
- description
- action_taken
- status
```

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10. HEALTH RECORDS

10.1 Features

- Medical history
- Allergies
- Clinic visits
- Emergency contacts

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11. DOCUMENT MANAGEMENT

11.1 Features

- Upload:
  - Birth certificate
  - Report cards
  - ID documents
- Secure storage
- Version control

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12. PARENT LINKING (CRITICAL)

12.1 Features

- Link one or multiple parents
- Multi-child support
- Secure access mapping

12.2 Flow

```
Student created
  ↓
Parent account linked
  ↓
Parent gains portal access
```

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13. COMMUNICATION INTEGRATION

13.1 Features

- Parent-teacher messaging
- Announcements
- Alerts

13.2 Triggers

- Absence alerts
- Grade updates
- Behavior incidents

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14. ANALYTICS & REPORTS

14.1 Dashboards

- Attendance trends
- Academic performance
- Behavior patterns

14.2 Advanced Insights

- At-risk students
- Top performers
- Class comparisons

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15. ALERTS & NOTIFICATIONS

15.1 Triggers

- Low attendance
- Poor academic performance
- Behavior incidents
- Fee issues (via finance module)

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16. SECURITY & ROLE-BASED ACCESS

16.1 Roles

Role	Access
Admin	Full access
Teacher	Assigned students only
Parent	Own child only
Student	Limited (optional portal)

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17. MOBILE-FIRST DESIGN

17.1 Requirements

- Teacher-friendly mobile attendance
- Parent mobile portal
- Offline capability

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18. SCALABILITY

- Support thousands of students
- Multi-campus support
- Fast search and filtering